

Prayas 2.0 2027 (OG)

Maths

Basic Mathematics

DPP: 3

Total Duration - 31 Min

Q1 The solution set of the inequality $\log_{\sin(\frac{\pi}{3})}(x^2 - 3x + 2) \geq 2$ is

- (A) $(\frac{1}{2}, 2)$
 (B) $(1, \frac{5}{2})$
 (C) $[\frac{1}{2}, 1) \cup (2, \frac{5}{2}]$
 (D) $(1, 2)$

Q2 If $\log_{0.3}(x - 1) < \log_{0.09}(x - 1)$, then x lies in the interval

- (A) $(2, \infty)$
 (B) $(1, 2)$
 (C) $(-2, -1)$
 (D) $(1, \frac{3}{2})$

Q3 Solution set of the inequality $2 - \log_2(x^2 + 3x) \geq 0$ is:

- (A) $[-4, 1]$
 (B) $[-4, -3) \cup (0, 1]$
 (C) $(-\infty, -3) \cup (1, \infty)$
 (D) $(-\infty, -4) \cup [1, \infty)$

Q4 If $\log_{0.5} \log_5(x^2 - 4) > \log_{0.5} 1$, then x lies in the interval

- (A) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 3)$
 (B) $(-3, -\sqrt{5}) \cup (\sqrt{5}, 2)$
 (C) $(\sqrt{5}, 3\sqrt{5})$
 (D) ϕ

Q5 Find the square of the sum of the roots of the equation

$$\log_3 x \cdot \log_4 x \cdot \log_5 x = \log_3 x \cdot \log_4 x + \log_4 x \cdot \log_5 x + \log_5 x \cdot \log_3 x$$

Q6 Solve the equation for x :
 $x^{\log x + 4} = 32$, where base of logarithm is 2.

Q7 Solve for
 $x : \log^2(4 - x) + \log(4 - x) \cdot \log(x + \frac{1}{2}) - 2 \log^2(x + \frac{1}{2}) = 0.$

Q8 The sum of all the solutions to the equation $2 \log_{10} x - \log_{10}(2x - 75) = 2$
 (A) 30 (B) 350
 (C) 75 (D) 200

Q9 Number of integral values of x the inequality $\log_{10}\left(\frac{2x-2007}{x+1}\right) \leq 0$ holds true, is

- (A) 1004 (B) 1005
 (C) 2007 (D) 2008

Q10 The product of all values of x satisfying the equation $|x - 1|^{\log_3 x^2 - 2 \log_x 9} = (x - 1)^7$, is:
 (A) 162
 (B) $\frac{162}{\sqrt{3}}$
 (C) $\frac{81}{\sqrt{3}}$
 (D) 81



Answer Key

Q1 (C)
Q2 (A)
Q3 (B)
Q4 (A)
Q5 3721

Q6 $1/32, 2$
Q7 $x = 0, \frac{7}{4}, \frac{3+\sqrt{24}}{2}$
Q8 (D)
Q9 (B)
Q10 (A)



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